

**UNITED STATES
SECURITIES AND EXCHANGE COMMISSION**

Washington, D.C. 20549

FORM 10-Q

(Mark One)

☒ **QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the quarterly period ended March 31, 2023

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to

Commission file number: 001-36481

ASPEN AEROGELS, INC.

(Exact name of registrant as specified in its charter)

Delaware

(State or other jurisdiction of
incorporation or organization)

30 Forbes Road, Building B
Northborough, Massachusetts
(Address of principal executive offices)

04-3559972

(I.R.S. Employer
Identification No.)

01532

(Zip Code)

Registrant's telephone number, including area code: (508) 691-1111

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol	Name of exchange on which registered
Common Stock, par value \$0.00001 per share	ASPN	The New York Stock Exchange

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days.
Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☐	Accelerated filer	☒
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of May 4, 2023, the registrant had 70,110,751 shares of common stock outstanding.

ASPEN AEROGELS, INC.

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Trademarks, Trade Names and Service Marks

We own or have rights to use “Aspen Aerogels,” “Cryogel,” “Pyrogel,” “Spaceloft,” “PyroThin,” the Aspen Aerogels logo and other trademarks, service marks and trade names of Aspen Aerogels, Inc. appearing in this Quarterly Report on Form 10-Q. Solely for convenience, the trademarks, service marks and trade names referred to in this report are presented without the ® and TM symbols, but such references are not intended to indicate, in any way, that the owner thereof will not assert, to the fullest extent under applicable law, such owner’s rights to these trademarks, service marks and trade names. This report contains additional trademarks, service marks and trade names of other companies, which, to our knowledge, are the property of their respective owners.

PART I — FINANCIAL INFORMATION

Item 1. Financial Statements.

ASPEN AEROGELS, INC.
Consolidated Balance Sheets
(Unaudited)

	March 31, 2023	December 31, 2022
	(In thousands, except share and per share data)	
Assets		
Current assets:		
Cash and cash equivalents	\$ 207,539	\$ 281,335
Restricted cash	629	1,226
Accounts receivable, net of allowances of \$346 and \$255	47,280	57,350
Inventories	27,007	22,538
Prepaid expenses and other current assets	13,581	7,236
Total current assets	296,036	369,685
Property, plant and equipment, net	312,068	259,223
Operating lease right-of-use assets	10,218	11,990
Other long-term assets	2,410	2,518
Total assets	\$ 620,732	\$ 643,416
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	\$ 52,822	\$ 54,728
Accrued expenses	9,763	16,003
Deferred revenue	3,974	5,846
Operating lease liabilities	2,267	2,368
Total current liabilities	68,826	78,945
Convertible note - related party	106,416	103,580
Operating lease liabilities long-term	12,948	13,456
Total liabilities	188,190	195,981
Commitments and contingencies (Note 9)		
Stockholders' equity:		
Preferred stock, \$0.00001 par value; 5,000,000 shares authorized, no shares issued and outstanding at March 31, 2023 and December 31, 2022	—	—
Common stock, \$0.00001 par value; 125,000,000 shares authorized, 70,069,160 and 69,994,963 shares issued and outstanding at March 31, 2023 and December 31, 2022, respectively	—	—
Additional paid-in capital	1,077,129	1,075,226
Accumulated deficit	(644,587)	(627,791)
Total stockholders' equity	432,542	447,435
Total liabilities and stockholders' equity	\$ 620,732	\$ 643,416

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Operations
(Unaudited)

	Three Months Ended March 31,	
	2023	2022
	(In thousands, except share and per share data)	
Revenue	\$ 45,586	\$ 38,407
Cost of revenue	40,500	40,195
Gross profit (loss)	5,086	(1,788)
Operating expenses:		
Research and development	4,099	3,592
Sales and marketing	7,713	6,018
General and administrative	12,182	7,226
Total operating expenses	23,994	16,836
Loss from operations	(18,908)	(18,624)
Other income (expense)		
Interest expense, convertible note - related party	(275)	(819)
Interest income (expense), net	2,387	(41)
Total other income (expense), net	2,112	(860)
Net loss	\$ (16,796)	\$ (19,484)
Net loss per share:		
Basic and diluted	\$ (0.24)	\$ (0.59)
Weighted-average common shares outstanding:		
Basic and diluted	69,162,739	32,940,040

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Stockholders' Equity
(Unaudited)
(In thousands, except share data)

	Preferred Stock \$0.00001 Par Value		Common Stock \$0.00001 Par Value		Additional Paid-in Capital	Accumulate d Deficit	Total Stockholder s' Equity
	Shares	Value	Shares	Value			
Balance at December 31, 2022	—	\$ —	69,994,963	\$ —	\$ 1,075,226	\$ (627,791)	\$ 447,435
Net loss	—	—	—	—	—	(16,796)	(16,796)
Stock compensation expense	—	—	—	—	2,267	—	2,267
Vesting of restricted stock units	—	—	71,643	—	(385)	—	(385)
Proceeds from employee stock option exercises	—	—	2,554	—	21	—	21
Balance at March 31, 2023	—	\$ —	70,069,160	\$ —	\$ 1,077,129	\$ (644,587)	\$ 432,542
	Preferred Stock \$0.00001 Par Value		Common Stock \$0.00001 Par Value		Additional Paid-in Capital	Accumulate d Deficit	Total Stockholder s' Equity
	Shares	Value	Shares	Value			
Balance at December 31, 2021	—	\$ —	33,218,115	\$ —	\$ 673,461	\$ (545,053)	\$ 128,408
Net loss	—	—	—	—	—	(19,484)	(19,484)
Stock compensation expense	—	—	—	—	1,828	—	1,828
Vesting of restricted stock units	—	—	166,211	—	(2,315)	—	(2,315)
Proceeds from employee stock option exercises	—	—	4,681	—	38	—	38
Proceeds from at-the-market offering, net of commissions of \$729 and issuance costs of \$318	—	—	737,288	—	23,272	—	23,272
Proceeds from private placement of common stock, net of fees and issuance costs of \$136	—	—	1,791,986	—	49,864	—	49,864
Balance at March 31, 2022	—	\$ —	35,918,281	\$ —	\$ 746,148	\$ (564,537)	\$ 181,611

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Cash Flows
(Unaudited)

	Three Months Ended March 31,	
	2023	2022
	(In thousands)	
Cash flows from operating activities:		
Net loss	\$ (16,796)	\$ (19,484)
Adjustments to reconcile net loss to net cash used in operating activities:		
Depreciation	2,704	2,128
Accretion of interest on convertible note - related party	—	819
Amortization of convertible note issuance costs	9	4
Amortization of debt discount due to modification of convertible note – related party	266	—
Provision for bad debt	99	(5)
Stock-compensation expense	2,267	1,828
Reduction in the carrying amount of operating lease right-of-use assets	632	570
Changes in operating assets and liabilities:		
Accounts receivable	9,971	(4,102)
Inventories	(4,469)	(3,518)
Prepaid expenses and other assets	(5,097)	(374)
Accounts payable	(5,516)	3,063
Accrued expenses	(6,240)	(3,082)
Deferred revenue	(1,872)	(120)
Operating lease liabilities	(609)	(556)
Net cash used in operating activities	(24,651)	(22,829)
Cash flows from investing activities:		
Capital expenditures	(49,378)	(14,504)
Net cash used in investing activities	(49,378)	(14,504)
Cash flows from financing activities:		
Proceeds from issuance of convertible note related party	—	100,000
Issuance costs from convertible note	—	(185)
Proceeds from employee stock option exercises	21	38
Payments made for employee restricted stock tax withholdings	(385)	(2,315)
Proceeds from at-the-market offering, net of commissions of \$729	—	23,590
Fees and issuance costs from at-the-market offering	—	(318)
Proceeds from private placement of common stock	—	50,000
Fees and issuance costs from private placement of common stock	—	(136)
Repayment of prepayment liability	—	(4,728)
Net cash provided by financing activities	(364)	165,946
Net increase in cash, cash equivalents and restricted cash	(74,393)	128,613
Cash, cash equivalents and restricted cash at beginning of period	282,561	76,564
Cash, cash equivalents, restricted cash and restricted cash equivalents at end of period	\$ 208,168	\$ 205,177
Supplemental disclosures of cash flow information:		
Interest paid	\$ —	\$ 52
Supplemental disclosures of non-cash activities:		
Right-of-use assets obtained in exchange for new operating lease liabilities	\$ —	\$ 68
Capitalized interest	\$ 2,561	\$ —
Changes in accrued capital expenditures	\$ 3,610	\$ 9,566

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS
(UNAUDITED)

(1) Description of Business and Basis of Presentation

Nature of Business

Aspen Aerogels, Inc. (the Company) is an aerogel technology company that designs, develops and manufactures innovative, high-performance aerogel insulation used primarily in the energy industrial and sustainable insulation materials markets. In addition, the Company has introduced a line of aerogel thermal barriers for use in battery packs in the electric vehicle market. The Company is also developing applications for its aerogel technology in the battery materials and a number of other high-potential markets.

The Company maintains its corporate offices in Northborough, Massachusetts. The Company has three wholly owned subsidiaries: Aspen Aerogels Rhode Island, LLC, Aspen Aerogels Germany, GmbH and Aspen Aerogels Georgia, LLC.

Liquidity

During the three months ended March 31, 2023, the Company incurred a net loss of \$16.8 million, used \$24.7 million of cash in operations and used \$49.4 million of cash for capital expenditures. The Company had unrestricted cash and cash equivalents of \$207.5 million as of March 31, 2023.

The Company is increasing investment in the research and development of next-generation aerogel products and manufacturing process technologies. In addition, the Company has developed a number of promising aerogel products and technologies for the electric vehicle market. The Company believes that the commercial potential for the Company's products and technology in the electric vehicle market is significant. Accordingly, the Company is hiring additional personnel, incurring additional operating expenses, and incurring significant capital expenditures to expand silica aerogel manufacturing capacity, build an automated thermal barrier fabrication operation, enhance research and development laboratory facilities and equipment, and construct a battery materials facility, among other efforts.

The Company expects its existing cash balance will be sufficient to support current operating requirements, current research and development activities and the capital expenditures required to support the evolving commercial opportunity in the electric vehicle market and other strategic business initiatives. However, the Company plans to supplement its cash balance with equity financings, debt financings, customer prepayments, or government grant and loan programs to provide the additional capital necessary to purchase the capital equipment, construct the new facilities, establish the operations and complete the aerogel capacity expansions required to support these evolving commercial opportunities and strategic business initiatives.

Unaudited Interim Financial Information

The accompanying unaudited interim consolidated financial statements include the accounts of the Company and have been prepared in accordance with accounting principles generally accepted in the United States of America (U.S. GAAP) and with the instructions to Form 10-Q and Article 10 of Regulation S-X. Certain information and disclosures normally included in the consolidated financial statements prepared in accordance with U.S. GAAP have been condensed or omitted pursuant to such rules and regulations. As such, the information included in this Quarterly Report on Form 10-Q should be read in conjunction with the audited consolidated financial statements and accompanying notes in our Annual Report on Form 10-K for the year ended December 31, 2022 (the Annual Report), filed with the U.S. Securities and Exchange Commission on March 16, 2023.

In the opinion of the Company's management, the unaudited interim consolidated financial statements have been prepared on the same basis as the audited consolidated financial statements and include all adjustments that are of a normal recurring nature and necessary for the fair statement of the Company's financial position as of March 31, 2023 and the results of its operations and stockholders' equity for the three months ended March 31, 2023 and 2022 and the cash flows for the three-month periods then ended. The Company has evaluated subsequent events through the date of this filing.

The Company's results of operations for the three months ended March 31, 2023 are not necessarily indicative of the results to be expected for the year ending December 31, 2023 or any other period.

(2) Significant Accounting Policies

Principles of Consolidation

The accompanying condensed consolidated financial statements of the Company include the accounts of all its subsidiaries which are majority-owned, controlled by the Company or a variable interest entity ("VIE") where the Company is the primary beneficiary. All significant intercompany accounts and transactions have been eliminated in consolidation.

A VIE is an entity that lacks one or more of the characteristics of a voting interest entity. A VIE is defined as an entity in which equity investors do not have the characteristics of a controlling financial interest or do not have sufficient equity at risk for the entity to finance its activities without additional subordinated financial support from other parties. The Company consolidates a VIE in accordance with ASC 810, Consolidation ("ASC 810") when it is the primary beneficiary of such VIE. As primary beneficiary, the Company has both the power to direct the activities that most significantly impact the economic performance of the VIE and a right to receive benefits or absorb losses of the entity that could be potentially significant to the VIE. The Company is required to reconsider its evaluation of whether to consolidate a VIE each reporting period, based upon changes in the facts and circumstances pertaining to the VIE.

The Company evaluates the initial consolidation of each Consolidated VIE, which includes a determination of whether the VIE constitutes the definition of a business in accordance with ASC 805, Business Combinations ("ASC 805"), by considering if substantially all of the fair value of the gross assets within the VIE are concentrated in either a single identifiable asset or group of single identifiable assets. Upon consolidation, the Company recognizes the assets acquired, the liabilities assumed, and any third-party ownership of membership interests as non-controlling interest as of the consolidation or acquisition date, measured at their relative fair values.

In April 2022, the Company engaged Prodensa Servicios de Consultora to establish OPE Manufacturer Mexico S de RL de CV, a maquiladora located in Mexico with the express purposes of manufacturing thermal barrier PyroThin products and ultimately constructing an automated fabrication facility for PyroThin. OPE is currently owned by Prodensa, which charges a management fee though there is an option for OPE to be purchased by the Company after a period of 18 months. During the period between inception and the purchase option, OPE operations are consolidated within the Company financial statements as of and for the quarter ended March 31, 2023.

Use of Estimates

The preparation of the consolidated financial statements requires the Company to make a number of estimates and assumptions that affect the reported amounts of assets and liabilities at the date of the consolidated financial statements and the reported amounts of revenues and expenses during the reporting period. Significant items subject to such estimates and assumptions include allowances for doubtful accounts, sales returns and allowances, product warranty costs, inventory valuation, the carrying amount of property and equipment, right-of-use assets, lease liabilities, stock-based compensation, and deferred income taxes. The Company evaluates its estimates and assumptions on an on-going basis using historical experience and other factors, including current economic conditions, which are believed to be reasonable under the circumstances. Management adjusts such estimates and assumptions when facts and circumstances warrant. Illiquid credit markets, volatile equity markets and declines in business investment can increase the uncertainty inherent in such estimates and assumptions. As future events and their effects cannot be determined with precision, actual results could differ significantly from these estimates. Changes in these estimates resulting from continuing changes in the economic environment will be reflected in the financial statements in future periods.

Cash and Cash Equivalents

Cash equivalents include short-term, highly liquid instruments, which consist of money market accounts and high-quality debt securities issued by the U.S. government via cash sweep accounts. All cash and cash equivalents are maintained with major financial institutions in North America. Deposits with these financial institutions may exceed the amount of insurance provided on such deposits; however, these deposits typically may be redeemed upon demand and, therefore, bear minimal risk.

In light of the recent failure of Silicon Valley Bank and other regional banks, the Company continues to establish commercial banking relationships with additional large financial institutions.

Restricted Cash

As of March 31, 2023, the Company had \$0.6 million of restricted cash to support its outstanding letters of credit to secure obligations under certain commercial contracts and other obligations.

Concentration of Credit Risk

Financial instruments, which potentially expose the Company to concentrations of credit risk, consist principally of accounts receivable. The Company's customers are primarily insulation distributors, insulation contractors, insulation fabricators and select energy and automotive end-users located throughout the world. The Company performs ongoing credit evaluations of its customers' financial condition and generally requires no collateral to secure accounts receivable. The Company maintains an allowance for doubtful accounts based on its assessment of the collectability of accounts receivable. The Company reviews the allowance for doubtful accounts quarterly. During the three months ended March 31, 2023, the Company recorded an increase for estimated customer uncollectible accounts receivable of less than \$0.1 million. During the three months ended March 31, 2022, the Company recorded a reduction for estimated customer uncollectible accounts receivable of less than \$0.1 million.

Revenue Recognition

The Company recognizes revenue in accordance with Accounting Standards Codification 606, Revenue from Contracts with Customers (ASC 606). See note 3 for further details.

Stock-based Compensation

The Company grants share-based awards to its employees and non-employee directors. All share-based awards granted, including grants of stock options, restricted stock and restricted stock units (RSUs), are recognized in the statement of operations based on their fair value as of the date of grant. Expense is recognized on a straight-line basis over the requisite service period for all awards with service conditions. For performance-based awards, the grant date fair value is recognized as expense when the condition is probable of being achieved, and then on a graded basis over the requisite service period. The Company uses the Black-Scholes option-pricing model to determine the fair value of service-based option awards. The Black-Scholes model requires the use of a number of complex and subjective assumptions including fair value of the underlying security, the expected volatility of the underlying security, a risk-free interest rate and the expected term of the option.

The fair value of restricted stock and RSUs is determined using the closing price of the Company's common stock on the date of grant. All shares of restricted stock are not transferable until vested. Restricted stock is typically issued to non-employee directors and typically vests over a one-year period from the date of issuance. RSUs are issued to employees and typically vest over a three-year period from the date of issuance. The fair value of restricted stock and RSUs upon which vesting is solely service-based is expensed ratably over the vesting period. If the service condition for shares of restricted stock is not met for any reason, the shares of unvested restricted stock will be forfeited and returned to the Company.

For stock options that contain a market condition, the Company uses the Monte-Carlo simulation option-pricing model to determine the fair value of the awards. In addition to the input assumptions used in the Black-Scholes model, the Monte-Carlo simulation option-pricing model factors the probability that the specific market condition may or may not be satisfied into the valuation. Stock-based compensation expense for awards with a market condition is recognized on a straight-line basis over the requisite service period for each such award.

During the three months ended March 31, 2023, the Company granted 326,642 restricted common stock units (RSUs) with an aggregate grant date fair value of \$3.2 million and non-qualified stock options (NSOs) to purchase 501,472 shares of common stock with an aggregate grant date fair value of \$3.2 million to employees under the 2014 Employee, Director, and Consultant Equity Incentive Plan (the 2014 Equity Plan). The RSUs and NSOs granted to employees will vest over a three-year period.

Stock-based compensation is included in cost of revenue or operating expenses, as applicable, and consists of the following:

	Three Months Ended March 31,	
	2023	2022
	(In thousands)	
Cost of product revenue	\$ 134	\$ 156
Research and development expenses	30	224
Sales and marketing expenses	314	324
General and administrative expenses	1,789	1,124
Total stock-based compensation	<u>\$ 2,267</u>	<u>\$ 1,828</u>

Pursuant to the “evergreen” provisions of the 2014 Equity Plan, the number of shares of common stock authorized for issuance under the plan automatically increased by 926,219 shares to 10,121,994 shares effective January 1, 2023.

As of March 31, 2023, 4,789,058 shares of common stock were reserved for issuance upon the exercise or vesting of outstanding stock-based awards granted under the 2014 Equity Plan and 2001 Equity Incentive Plan, as amended (the 2001 Equity Plan). Any cancellations or forfeitures of the options outstanding under the 2001 Equity Plan will result in the shares reserved for issuance upon exercise of such options becoming available for grant under the 2014 Equity Plan. As of March 31, 2023, the Company has either reserved in connection with statutory tax withholdings or issued a total of 4,797,111 shares under the 2014 Equity Plan. As of March 31, 2023, there were 535,825 shares of common stock available for future grant under the 2014 Equity Plan.

Net Loss per Share

The Company calculates net loss per share of common stock based on the weighted-average number of shares of common stock outstanding during each period. Potential common stock equivalents are determined using the treasury stock method. The weighted-average number of shares of common stock included in the computation of diluted net loss gives effect to all potentially dilutive common equivalent shares, including outstanding stock options and RSUs. Common equivalent shares are excluded from the computation of diluted net loss per share if their effect is antidilutive.

Warranty

The Company provides warranties for its products and records the estimated cost within cost of revenue in the period that the related revenue is recorded.

The Company’s standard warranty period for energy industrial products extends to one year from the date of shipment. This standard warranty provides that the Company’s products will be free from defects in material and workmanship, and will, under normal use, conform to the specifications for the product.

The Company’s thermal barrier products provide quality and warranty provisions customary in the automotive industry.

The Company recorded warranty expense related to its thermal barrier products of less than \$0.1 million during each of three months ended March 31, 2023 and 2022.

Recently Issued Accounting Standards

From time to time, new accounting pronouncements are issued by the Financial Accounting Standards Board or other standard setting bodies. Recently issued standards typically do not require adoption until a future effective date. Prior to their effective date, the Company evaluates the pronouncements to determine the potential effects of adoption to its consolidated financial statements.

Standards Implemented Since December 31, 2022

The Company has not implemented any accounting standards that had a material impact on its consolidated financial statements during the three months ended March 31, 2023.

Standards to be Implemented

The Company believes that the impact of recently issued accounting standards that are not yet effective will not have a material impact on its consolidated financial statements.

(3) Revenue from Contracts with Customers

Revenue Recognition

Revenue is recognized when a customer obtains control of promised goods or services in an amount that reflects the consideration which the entity expects to receive in exchange for those goods or services. To determine revenue recognition for arrangements within the scope of ASC 606, the Company performs the following five steps: (i) identification of the contract with a customer; (ii) identification of the performance obligations in the contract; (iii) determination of the transaction price; (iv) allocation of the transaction price to the separate performance obligations in the contract; and (v) recognition of the revenue associated with performance obligations as they are satisfied. The Company applies the five-step model to contracts when it is probable that the Company will collect the consideration it is entitled to in exchange for the goods or services it transfers to the customer. At contract inception, once the contract is determined to be within the scope of ASC 606, the Company assesses the goods or services promised within each contract and determines those that are performance obligations and assesses whether each promised good or service is distinct. If the contract contains a single performance obligation, the entire transaction price is allocated to the single performance obligation. Contracts that contain multiple performance obligations require an allocation of the transaction price based on the estimated relative standalone-selling prices of the promised products or services underlying each performance obligation. The Company determines standalone-selling prices based on the price at which the performance obligation is sold separately. If the standalone-selling price is not observable through past transactions, the Company estimates the standalone-selling price considering available information such as market conditions and internally approved pricing guidelines related to the performance obligations. The Company then recognizes as revenue the amount of the transaction price that is allocated to the respective performance obligation when (or as) the performance obligation is satisfied.

When determining the transaction price of a contract, an adjustment is made if payment from a customer occurs either significantly before or significantly after performance, resulting in a significant financing component. Applying the practical expedient in paragraph ASC 606-10-32-18, the Company does not assess whether a significant financing component exists if the period between when the Company performs its obligations under the contract and when the customer pays is one year or less. The Company did not have any contracts outstanding at December 31, 2022 and did not enter into any contracts during the three months ended March 31, 2023 that contained a significant financing component.

The Company records deferred revenue for product sales when (i) the Company has delivered products, but other revenue recognition criteria have not been satisfied, or (ii) payments have been received in advance of the completion of required performance obligations.

Shipping and Handling Costs

Shipping and handling costs associated with outbound freight after control over a product has transferred to a customer are accounted for as fulfillment costs and are included in the cost of product revenue. The associated amount of revenue recognized includes the consideration to which the Company expects to be entitled to receive in exchange for incurring these shipping and handling costs.

Energy Industrial

The Company generally enters into contracts containing one type of performance obligation. The Company recognizes revenue when the performance obligation is satisfied, which is generally upon delivery according to contractual shipping terms within customer purchase orders.

The Company also enters into rebate agreements with certain customers. These agreements may be considered an additional performance obligation of the Company or variable consideration within a contract. Rebates are recorded as a reduction of revenue in the period the related revenue is recognized. A corresponding liability is recorded as a component of deferred revenue on the consolidated balance sheets. These arrangements are primarily based on the customer attaining contractually specified sales volumes.

The Company estimates the amount of its sales that may be returned by its customers and records this estimate as a reduction of revenue in the period the related revenue is recognized. The Company currently estimates return liabilities using historical rates of return, current quarter credit sales, and specific items of exposure on a contract-by-contract basis. Sales return reserves were approximately \$0.1 million at both March 31, 2023 and December 31, 2022.

The Company manufactures and sells subsea products that are designed for pipe-in-pipe applications in offshore oil production and are typically customized to meet customer specifications. Subsea products typically have no alternative use and contain an enforceable right to payment. Customer invoicing terms for subsea products are typically based on certain milestones within the production and delivery schedule. Under the provisions of ASC 606, the Company recognizes revenue at a point in time when transfer of control of the products is passed to the customer, or over time utilizing the input method. The timing of revenue recognition is assessed on a contract-by-contract basis.

Thermal Barriers

The Company supplies fabricated, multi-part thermal barriers for use in battery packs in the electric vehicle market. These thermal barriers are customized to meet customer specifications. Thermal barrier products typically have no alternative use and may contain an enforceable right to payment. Under the provisions of ASC 606, the Company may recognize revenue at a point in time when transfer of the control of the products is passed to the customer, or over time utilizing the input method. The timing of revenue recognition is assessed on a contract-by-contract basis. During the three months ended March 31, 2023 and 2022, the Company recognized revenue of \$11.7 million and \$7.6 million, respectively, from thermal barrier contracts.

Disaggregation of Revenue

In the following tables, revenue is disaggregated by primary geographical region and source of revenue:

	Three Months Ended March 31,					
	2023			2022		
	U.S.	International	Total	U.S.	International	Total
	(In thousands)					
Geographical region						
Asia	\$ —	\$ 11,784	\$ 11,784	\$ —	\$ 7,324	\$ 7,324
Canada	—	324	324	—	860	860
Europe	—	5,412	5,412	—	3,911	3,911
Latin America	—	1,624	1,624	—	1,606	1,606
U.S.	26,442	—	26,442	24,706	—	24,706
Total revenue	<u>\$ 26,442</u>	<u>\$ 19,144</u>	<u>\$ 45,586</u>	<u>\$ 24,706</u>	<u>\$ 13,701</u>	<u>\$ 38,407</u>
Source of revenue						
Energy industrial	\$ 16,504	\$ 17,371	\$ 33,875	\$ 18,275	\$ 12,500	\$ 30,775
Thermal barrier	9,938	1,773	11,711	6,431	1,201	7,632
Total revenue	<u>\$ 26,442</u>	<u>\$ 19,144</u>	<u>\$ 45,586</u>	<u>\$ 24,706</u>	<u>\$ 13,701</u>	<u>\$ 38,407</u>

Contract Balances

The following table presents changes in the Company's contract assets and contract liabilities during the three months ended March 31, 2023:

	Balance at December 31, 2022	Additions	Deductions	Balance at March 31, 2023
	(In thousands)			
Contract assets				
Thermal barrier	\$ 143	\$ —	\$ (143)	\$ —
Total contract assets	<u>\$ 143</u>	<u>\$ —</u>	<u>\$ (143)</u>	<u>\$ —</u>
Contract liabilities				
Deferred revenue				
Energy industrial	\$ 5,846	\$ 2,036	\$ (3,908)	\$ 3,974
Total contract liabilities	<u>\$ 5,846</u>	<u>\$ 2,036</u>	<u>\$ (3,908)</u>	<u>\$ 3,974</u>

During the three months ended March 31, 2023, the Company recognized \$2.8 million of revenue that was included in deferred revenue as of December 31, 2022.

A contract asset is recorded when the Company satisfies a performance obligation by transferring a promised good or service and has earned the right to consideration from its customer. These assets may represent a conditional right to consideration and are included within accounts receivable and other current assets on the consolidated balance sheets.

A contract liability is recorded when consideration is received, or such consideration is unconditionally due, from a customer prior to transferring goods or services under the terms of the contract. Contract liabilities are recognized as revenue after control of the products or services is transferred to the customer and all revenue recognition criteria have been met.

(4) Inventories

Inventories consist of the following:

	March 31, 2023	December 31, 2022
	(In thousands)	
Raw materials	\$ 21,735	\$ 19,877
Finished goods	5,272	2,661
Total	<u>\$ 27,007</u>	<u>\$ 22,538</u>

(5) Property, Plant and Equipment, Net

Property, plant and equipment consist of the following:

	March 31, 2023	December 31, 2022	Useful life
	(In thousands)		
Construction in progress	\$ 259,194	\$ 209,056	—
Buildings	24,016	24,016	30 years
Machinery and equipment	139,720	136,607	3-10 years
Computer equipment and software	10,333	10,239	3 years
Leasehold improvements	11,325	9,226	Shorter of useful life or lease term
Total	444,588	389,144	
Accumulated depreciation	(132,520)	(129,921)	
Property, plant and equipment, net	<u>\$ 312,068</u>	<u>\$ 259,223</u>	

Depreciation expense was \$2.7 million and \$2.1 million for the three months ended March 31, 2023 and 2022, respectively.

Construction in progress totaled \$259.2 million and \$209.1 million at March 31, 2023 and December 31, 2022, respectively. The balance at March 31, 2023 and December 31, 2022 included engineering designs and construction costs totaling \$204.2 million and \$164.5 million, respectively, for a planned aerogel manufacturing facility in Bulloch County, Georgia. Capitalized interest totaled \$2.6 million and \$2.7 million at March 31, 2023 and December 31, 2022, respectively.

(6) Accrued Expenses

Accrued expenses consist of the following:

	March 31, 2023	December 31, 2022
	(In thousands)	
Employee compensation	\$ 6,705	\$ 12,467
Other accrued expenses	3,058	3,536
Total	<u>\$ 9,763</u>	<u>\$ 16,003</u>

(7) Related Party Transactions

Convertible Note

During the year ended December 31, 2022, the Company issued a \$100.0 million aggregate principal amount convertible note to Wood River Capital, LLC, an entity affiliated with Koch Disruptive Technologies, LLC (the 2022 Convertible Note). Refer to note 8 for more information.

During the three months ended March 31, 2023, the Company incurred \$2.6 million of interest from the 2022 Convertible Note, and capitalized \$2.6 million as part of the construction in progress for the planned manufacturing facility in Bulloch County, Georgia.

On November 28, 2022, the Company entered into an amendment to the 2022 Convertible Note, or the Convertible Note Amendment, to reduce the initial Conversion Price (as defined in the 2022 Convertible Note) by \$5.00 per share from \$34.936625 per share to \$29.936625 per share, by increasing the initial Conversion Rate (as defined in the 2022 Convertible Note) from 28.623257 shares per \$1,000 of Capitalized Principal Amount (as defined in the 2022 Convertible Note) to 33.400100 shares per \$1,000 of Capitalized Principal Amount under the 2022 Convertible Note, subject to customary anti-dilution and other adjustments (as described in the Indenture, which governs the 2022 Convertible Note).

Other

During the three months ended March 31, 2023, the Company recorded costs of \$4.6 million as a component of construction in progress in connection with the planned aerogel manufacturing facility in Bulloch County, Georgia in fees from Koch Project Solutions, LLC, an entity affiliated with Koch Disruptive Technologies, LLC for project management service.

(8) Convertible Note – Related Party

2022 Convertible Note

On February 15, 2022, the Company entered into a note purchase agreement (the Note Purchase Agreement) with Wood River Capital LLC, an entity affiliated with Koch Disruptive Technologies, LLC (Koch), relating to the issuance and sale to Koch of the 2022 Convertible Note in the aggregate principal amount of \$100.0 million. The transactions contemplated by the Note Purchase Agreement closed on February 18, 2022 (the Issue Date). The maturity date of the 2022 Convertible Note is February 18, 2027, subject to earlier conversion, redemption, or repurchase.

The 2022 Convertible Note is a senior unsecured obligation of the Company and ranks equal in right of payment to all senior unsecured indebtedness of the Company, and will rank senior in right of payment to any indebtedness that is contractually subordinated to the 2022 Convertible Note.

In accordance with ASU 2020-06, the 2022 Convertible Note is accounted for as a single unit of account and consists of the following:

	March 31, 2023
Convertible note, principal	\$ 100,000
Payment in-kind	7,830
Accrued interest	2,561
Discount on convertible note, net of accumulated amortization	(3,831)
Debt issuance costs, net of accumulated amortization	(144)
Convertible note	<u>\$ 106,416</u>

In general, fair values determined by Level 1 inputs utilize observable inputs such as quoted prices in active markets for identical assets or liabilities. Fair values determined by Level 2 inputs utilize data points that are either directly or indirectly observable, such as quoted prices for similar instruments in active markets, interest rates and yield curves. Fair values determined by Level 3 inputs utilize unobservable data points in which there is little or no market data, which require the Company to develop its own assumptions for the asset or liability. The 2022 Convertible Note does not have current observable inputs such as recent trading prices (Level 3) and is measured at fair value using a combination of option pricing and discounted cash flow models and incorporate management's assumptions for stock price, volatility and risk rate.

The Company estimated the fair value of the 2022 Convertible Notes is approximately \$78.4 million as of March 31, 2023. However, as the Company has not elected to utilize the fair value option, it is carried at amortized cost of \$106.4 million.

Contractual Interest Rates

The 2022 Convertible Note was issued at par and bears interest at the Secured Overnight Financing Rate (SOFR) plus 5.50% per annum if interest is paid in cash, or, if interest is paid in-kind as an increase in the principal amount of the outstanding note, at the SOFR plus 6.50% per annum. Under the terms of the 2022 Convertible Note, SOFR has a floor of 1% and a cap of 3%. Interest on the 2022 Convertible Note is payable semi-annually in arrears on June 30 and December 30. The Company, at its option, is permitted to settle each semi-annual interest payment in cash, in-kind, or any combination thereof. It is expected that the Notes will mature on February 18, 2027, subject to earlier conversion, redemption or repurchase.

The Company elected to repay the contractual interest due on June 30, 2022 and December 30, 2022 in-kind as an increase to the principal amount of \$2.9 million and \$4.9 million, respectively. The contractual interest attributable to the 2022 Convertible Note was recorded as an addition to the convertible note – related party balance on the condensed consolidated balance sheets.

Accrued interest was \$2.6 million as of March 31, 2023, of which debt issuance costs comprised of \$0.1 million. The effective interest rate approximated the contract interest rate for the three months ended March 31, 2023. The Company amortized \$0.3 million of the \$4.1 million discount on the convertible note as of March 31, 2023 utilizing an effective interest rate of 10.7%.

Conversion Rights

On November 28, 2022, the Company entered into an amendment to the 2022 Convertible Note to reduce the initial Conversion Price by \$5.00 per share from \$34.936625 per share to \$29.936625 per share, by increasing the initial Conversion Rate from 28.623257 shares per \$1,000 of Capitalized Principal Amount to 33.400100 shares per \$1,000 of Capitalized Principal Amount under the Convertible Note. Accordingly, the 2022 Convertible Note is convertible at the option of the holder at any time prior to the business day immediately preceding the maturity date at an initial conversion rate of 33.400100 shares of the Company's common stock per \$1,000 of capitalized principal. The effective conversion price is approximately \$29.936625 per share (the Conversion Price). The Conversion Price is subject to adjustment upon the occurrence of certain dilutive events such as stock splits and combinations, stock dividends, mergers and spin-off. As of March 31, 2023, 3,687,070 shares of the Company's common stock were issuable upon conversion of the 2022 Convertible Note. The Company has the right to settle conversions in shares of common stock, cash, or any combination thereof. If the closing price per share of the Company's common stock on the New York Stock Exchange is at least 130% of the Conversion Price for 20 consecutive trading days, the Company may elect to convert the principal and accrued interest owing under the Notes, plus a make-whole amount equal to the sum of the present values of the remaining interest payments that would have otherwise been payable from the date of such conversion, redemption or repurchase, as applicable, through maturity (the Make-Whole Amount), into the Company's common stock at the Conversion Price.

Optional Redemption

The 2022 Convertible Note is redeemable at the Company's option at any time and in the event that the volume weighted average price of the Company's common stock for the 10 trading days immediately preceding the date on which the Company provides the redemption notice has been at least 130% of the Conversion Price then in effect at a redemption price of 100% of the principal amount, plus accrued and unpaid interest (excluding the redemption date), plus the Make-Whole Amount.

Contingent Redemption

Upon the occurrence of certain fundamental changes described in the Indenture (each, a Fundamental Change), the Holder of the Note may require that the Company repurchase all or part of the principal amount of the Note at a purchase price of 100% of the principal amount of such Note, plus accrued and unpaid interest to, but excluding, the Fundamental Change repurchase date, plus the Make-Whole Amount. The Indenture includes customary "events of default," which may result in the acceleration of the maturity of the Note.

Embedded Derivatives

The Company determined that the Make-Whole feature of the 2022 Convertible Note requires bifurcation in accordance with Accounting Standards Codification 815, Derivatives and Hedging (ASC 815). Accordingly, the Company must separately account for the feature at fair value with changes in fair value reported in current period earnings. The fair value of the Make-Whole was determined to be immaterial as of February 18, 2022 and March 31, 2023.

(9) Commitments and Contingencies

Cloud Computing Agreement

The Company is party to a cloud computing agreement that is a service contract for enterprise resource planning software. During the year ended December 31, 2022, the Company amended the agreement to a new five-year term. As of March 31, 2023, the Company had \$1.7 million of amortized costs related to implementation of the agreement that began to amortize during 2022. The capitalized implementation costs are classified on the consolidated balance sheets as follows:

	March 31, 2023	December 31, 2022
	(In thousands)	
Cloud computing costs included in other current assets	\$ 420	\$ 420
Cloud computing costs included in other assets	1,590	1,590
Amortization of cloud computing costs	(346)	(242)
Total capitalized cloud computing costs	<u>\$ 1,665</u>	<u>\$ 1,768</u>

Thermal Barrier Contracts

The Company is party to production contracts with General Motors to supply fabricated, multi-part thermal barriers (Barriers) for use in the battery system of its next-generation electric vehicles (Contracts). Pursuant to the Contracts, the Company is obligated to supply Barriers at fixed annual prices and at volumes to be specified by General Motors up to a daily maximum quantity through the respective terms of the agreements, which expire at various times from 2026 through 2034. While General Motors has agreed to purchase its requirement for Barriers from the Company for locations to be designated from time to time by General Motors, it has no obligation to purchase any minimum quantity of Barriers under the Contracts. In addition, General Motors may terminate the Contracts at any time and for any or no reason. All other terms of the Contracts are generally consistent with General Motors' standard purchase terms, including quality and warranty provisions customary in automotive industry.

Federal, State and Local Environmental Regulations

The Company is subject to federal, state and local environmental laws and regulations. These laws generally provide for control of pollutants released into the environment and require responsible parties to undertake remediation. Penalties may be imposed for noncompliance.

Litigation

The Company is, from time to time, a party to litigation that arises in the normal course of its business operations. See Part II, Item 1 “Legal Proceedings” of this Quarterly Report on Form 10-Q for a description of certain of the Company’s current legal proceedings. The Company is not presently a party to any litigation for which it believes a loss is probable requiring an amount to be accrued or a possible loss contingency requiring disclosure.

(10) Leases

The Company leases office, laboratory, warehouse and fabrication space in Massachusetts, Rhode Island and Monterrey, Mexico under operating leases. Under these agreements, the Company is obligated to pay annual rent, real estate taxes, and certain other operating expenses. The Company also leases equipment under operating leases. The Company’s operating leases expire at various dates through 2034.

The Company determines if an arrangement is a lease at inception. Right-of-use (ROU) assets represent the Company’s right to use an underlying asset for the lease term and lease liabilities represent the Company’s payment obligations under the lease. Operating lease ROU assets and liabilities are recognized based on the present value of lease payments over the lease term. To measure its lease liabilities, the Company uses its incremental borrowing rate or the rate implicit in the lease, if available. The Company calculates its incremental borrowing rate using a synthetic credit rating analysis based on Moody’s Building Materials Industry Rating Methodology. ROU assets also include any direct costs and prepaid lease payments but exclude any lease incentives received. The Company’s lease terms may include options to extend or terminate the lease when it is reasonably certain that the Company will exercise that option. Lease expense for lease payments is recognized on a straight-line basis over the lease term.

The Company elected the short-term lease recognition exemption for all leases that qualify. For leases that qualify for this exemption, the Company does not recognize ROU assets or lease liabilities. For lease agreements with lease and non-lease components, the Company accounts for each component separately. However, in the case of equipment leases, the Company accounts for lease and non-lease components as a single component.

Maturities of operating lease liabilities as of March 31, 2023 are as follows:

Year	Operating Leases (In thousands)
2023 (excluding the three months ended March 31, 2023)	\$ 2,888
2024	2,874
2025	2,638
2026	2,245
2027	1,899
Thereafter	12,042
Total lease payments	24,586
Less imputed interest	(9,371)
Total lease liabilities	\$ 15,215

The Company incurred operating lease costs of \$1.1 million and \$0.8 million during the three months ended March 31, 2023 and 2022, respectively. Cash payments related to operating lease liabilities were \$1.0 million and \$0.8 million during the three months ended March 31, 2023 and 2022, respectively.

As of March 31, 2023, the weighted average remaining lease term for operating leases was 9.2 years. As of March 31, 2023, the weighted average discount rate for operating leases was 11.3%.

As of March 31, 2023, the Company had additional real estate leases that will commence during 2023 with total lease payments of \$17.3 million and a weighted average lease term of 10.0 years.

As of March 31, 2023, the Company had no additional operating equipment leases that would commence during 2023.

(11) Net Loss Per Share

The computation of basic and diluted net loss per share consists of the following:

	Three Months Ended March 31,	
	2023	2022
(In thousands, except share and per share data)		
Numerator:		
Net loss	\$ (16,796)	\$ (19,484)
Denominator:		
Weighted average shares outstanding, basic and diluted	69,162,739	32,940,040
Net loss per share, basic and diluted	\$ (0.24)	\$ (0.59)

Potentially dilutive common shares that were excluded from the computation of diluted net loss per share because they were anti-dilutive consist of the following:

	Three Months Ended March 31,	
	2023	2022
Common stock options	4,330,797	3,952,596
Restricted common stock units	458,262	253,691
Restricted common stock awards	857,933	476,550
Convertible note, if converted	3,687,070	2,886,426
Total	9,334,062	7,569,263

As the Company incurred a net loss for the three months ended March 31, 2023 and 2022, the potential dilutive shares from common stock options, restricted common stock units, restricted common stock awards, and the convertible note were excluded from the calculation of diluted net loss per share because their effect would have been anti-dilutive for the periods presented. The Company excludes the shares issued in connection with restricted stock awards from the calculation of basic weighted average common shares outstanding until the restrictions lapse.

(12) Income Taxes

The Company incurred net operating losses and recorded a full valuation allowance against net deferred tax assets for all periods presented. Accordingly, the Company has not recorded a provision for federal or state income taxes.

(13) Segment Information

Operating segments are identified as components of an enterprise about which separate, discrete financial information is available for evaluation by the chief operating decision maker in making decisions on how to allocate resources and assess performance. The Company's chief operating decision maker is the Chief Executive Officer. The Company's chief operating decision maker reviews consolidated operating results to make decisions about allocating resources and assessing performance for the entire Company. The Company reports two segments: Energy Industrial and Thermal Barrier. We evaluate segment performance based on the segment profit (loss) before corporate expenses.

Summarized below are the Revenue and Segment Operating Profit for each reporting segment:

	Three Months Ended		Three Months Ended	
	Revenue		Segment Operating Profit (Loss)	
	March 31,		March 31,	
	2023	2022	2023	2022
	(In thousands)			
Energy industrial	\$ 33,875	\$ 30,775	\$ 8,881	\$ 2,997
Thermal barrier	11,711	7,632	(3,795)	(4,785)
Total	<u>\$ 45,586</u>	<u>\$ 38,407</u>	<u>\$ 5,086</u>	<u>\$ (1,788)</u>
Corporate expenses			23,994	16,836
Operating loss			(18,908)	(18,624)
Other income (expense), net			2,112	(860)
Net loss			<u>\$ (16,796)</u>	<u>\$ (19,484)</u>

	Total Assets	
	March 31, 2023	December 31, 2022
	(In thousands)	
Energy industrial	\$ 99,126	\$ 94,415
Thermal barrier	35,350	39,320
Total assets of reportable segments	134,476	133,735
Construction in progress	259,194	209,050
All other corporate assets	227,062	300,631
	<u>\$ 620,732</u>	<u>\$ 643,416</u>

(14) Subsequent Events

The Company has evaluated subsequent events through May 4, 2023, the date of issuance of the consolidated financial statements for the three months ended March 31, 2023.

Item 2. Management's Discussion and Analysis of Financial Condition and Results of Operations.

The following information should be read in conjunction with the unaudited financial information and the notes thereto included in this Quarterly Report on Form 10-Q and the audited financial information and the notes thereto included in the Annual Report on Form 10-K for the year ended December 31, 2022, filed with the U.S. Securities and Exchange Commission (SEC) on March 16, 2023, which we refer to as the Annual Report.

Certain matters discussed in this Quarterly Report on Form 10-Q may be deemed to be forward-looking statements that involve risks and uncertainties. We make such forward-looking statements pursuant to the safe harbor provisions of the Private Securities Litigation Reform Act of 1995 and other federal securities laws. In this Quarterly Report on Form 10-Q, words such as "may," "will," "anticipate," "estimate," "expects," "projects," "intends," "plans," "believes" and similar expressions (as well as other words or expressions referencing future events, conditions or circumstances) are intended to identify forward-looking statements.

Our actual results and the timing of certain events may differ materially from the results discussed, projected, anticipated, or indicated in any forward-looking statements. We caution you that forward-looking statements are not guarantees of future performance and that our actual results of operations, financial condition and liquidity, and the development of the industry in which we operate may differ materially from the forward-looking statements contained in this Quarterly Report on Form 10-Q. In addition, even if our results of operations, financial condition and liquidity, and the development of the industry in which we operate are consistent with the forward-looking statements contained in this Quarterly Report on Form 10-Q, they may not be predictive of results or developments in future periods.

The following information and any forward-looking statements should be considered in light of factors discussed elsewhere in this Quarterly Report on Form 10-Q and under "Risk Factors" in Item 1A of the Annual Report.

We caution readers not to place undue reliance on any forward-looking statements made by us, which speak only as of the date they are made. We disclaim any obligation, except as specifically required by law and the rules of the SEC, to publicly update or revise any such statements to reflect any change in our expectations or in events, conditions or circumstances on which any such statements may be based, or that may affect the likelihood that actual results will differ from those set forth in the forward-looking statements.

You should read the following discussion and analysis of financial condition and results of operations together with Part I Item 1 "Financial Statements," which includes our financial statements and related notes, elsewhere in this Quarterly Report on Form 10-Q.

Investors and others should note that we routinely use the Investors section of our website to announce material information to investors and the marketplace. While not all of the information that we post on the Investors section of our website is of a material nature, some information could be deemed to be material. Accordingly, we encourage investors, the media, and others interested in us to review the information that we share on the Investors section of our website, <https://www.aerogel.com>.

Products

Our core businesses are organized into two reportable segments: Energy Industrial and Thermal Barrier. The following describes our key product offerings and new product innovations by reportable segment.

Energy Industrial

We design, develop and manufacture innovative, high-performance aerogel insulation used primarily in the energy industrial and sustainable insulation materials markets. We believe our aerogel blankets deliver the best thermal performance of any widely used insulation product available on the market today and provide a combination of performance attributes unmatched by traditional insulation materials. Our end-user customers select our products where thermal performance is critical and to save money, improve resource efficiency, enhance sustainability, preserve operating assets and protect workers. Our insulation is used by oil producers and the owners and operators of refineries, petrochemical plants, liquefied natural gas facilities, power generating assets and other energy industrial. Our Pyrogel and Cryogel product lines have undergone rigorous technical validation by industry leading end-users and achieved significant market adoption. Our Spaceloft sustainable insulation materials are increasingly used by building owners to improve the energy efficiency and to enhance fire protection in buildings ranging from historic brownstones to modern high rises.

We also derive revenue from a number of other end markets. Customers in these markets use our products for applications as diverse as military and commercial aircraft, trains, buses, appliances, apparel, footwear and outdoor gear. As we continue to enhance our Aerogel Technology Platform, we believe we will have additional opportunities to address high-value applications in the global

insulation market, the electric vehicle market and in a number of new, high-value markets, including hydrogen energy, filtration, water purification, and gas sorption.

We market and sell our products primarily through a sales force based in North America, Europe and Asia. The efforts of our sales force are supported by a small number of sales consultants with extensive knowledge of a particular market or region. Our sales force is responsible for establishing and maintaining customer and partner relationships, delivering highly technical information and ensuring high-quality customer service.

Our salespeople work directly with end-user customers and engineering firms to promote the qualification, specification and acceptance of our aerogel and thermal barrier products. We also rely on an existing and well-established channel of qualified insulation distributors and contractors in more than 50 countries around the world to ensure rapid delivery of our aerogel products and strong end-user support.

Thermal Barrier

We are also actively developing a number of promising aerogel products and technologies for the electric vehicle market. We have developed and are commercializing our proprietary line of PyroThin aerogel thermal barriers for use in battery packs in electric vehicles. Our PyroThin product is an ultra-thin, lightweight and flexible thermal barrier designed with other functional layers to impede the propagation of thermal runaway across multiple lithium-ion battery system architectures. Our thermal barrier technology is designed to offer a unique combination of thermal management, mechanical performance and fire protection properties. These properties enable electric vehicle manufacturers to achieve critical battery performance and safety goals. In addition, we are seeking to leverage our patented carbon aerogel technology to develop industry-leading battery materials for use in lithium-ion battery cells.

These battery materials have the potential to increase the energy density of the battery cells, thus enabling an increase in the driving range of electric vehicles.

The commercial potential for our PyroThin thermal barriers and our carbon aerogel battery materials in the electric vehicle market is significant. Accordingly, we are hiring additional personnel, incurring additional operating expenses, incurring significant capital expenditures to expand aerogel manufacturing capacity, establishing an automated thermal barrier fabrication operation, enhancing research and development resources and expanding our battery material research facilities, among other items.

We have entered into production contracts with certain major OEMs, including General Motors LLC, or GM, to supply fabricated, multi-part thermal barriers for use in the battery system of its next-generation electric vehicles. Pursuant to the contracts with GM, we are obligated to supply the barriers at fixed annual prices and at volumes to be specified by the customer up to a daily maximum quantity through the term of the agreements, which expire at various times from 2026 through 2034. While GM has agreed to purchase its requirement for the barriers from us at locations to be designated from time to time, it has no obligation to purchase any minimum quantity of barriers under the contracts. In addition, GM may terminate the contracts any time and for any or no reason. All other terms of the contracts are generally consistent with GM's standard purchase terms, including quality and warranty provisions customary in the automotive industry.

Manufacturing Operations

We manufacture our products using our proprietary technology at our facility in East Providence, Rhode Island. We have operated the East Providence facility since 2008 and have increased our capacity in phases to approximately \$250.0 million in annual revenue. To meet expected growth in demand for our aerogel products in the electric vehicle market, we have been in the process of expanding our aerogel blanket capacity by constructing a second manufacturing plant in Bulloch County, Georgia. However, in order to manage the development of the second plant so that its increased capacity comes online in a manner that aligns with our current expectations as to demand from our EV customers, we are extending the timeframe for construction and commissioning of the second plant until such time as its capacity is supported by increased demand. In the meantime, and until we restart construction, we expect to be able to substantially reduce our planned capital expenditures for 2023 and 2024. At the same time, we believe that productivity improvements in our existing Rhode Island facility combined with supply of our energy industrial products from one or more contract manufacturers in China beginning in 2024 will permit us to achieve a target revenue capacity of approximately \$550.0 million in 2024 and prior to the completion and start-up of the second plant. Nonetheless, there can be no assurance as to when we will restart construction on the second plant and we are in the process of determining what remaining or additional capital expenditures and other costs may result from the decision to extend the timeframe for construction of the second plant. There can also be no assurance that our contract manufacturing strategy of meeting the demand of our energy industrial customers with supply from one or more contract manufacturers in China will provide us with adequate manufacturing capacity or supply for that expected demand. Furthermore, if and when we restart construction on the second plant, further cost inflation and/or supply chain disruptions, as well as potential changes in the scope of the facilities, could lead to increases to our prior estimates for completion of the second plant.

Financial Summary

Our revenue for the three months ended March 31, 2023 was \$45.6 million, which represented an increase of \$7.2 million, or 19%, from \$38.4 million for the three months ended March 31, 2022. Net loss for the three months ended March 31, 2023 was \$16.8 million and net loss per share was \$0.24. Net loss for the three months ended March 31, 2022 was \$19.5 million and net loss per share was \$0.59.

Key Metrics and Non-GAAP Financial Measures

We regularly review a number of metrics, including the following key metrics, to evaluate our business, measure our performance, identify trends affecting our business, formulate financial projections and make strategic decisions.

Square Foot Operating Metric

We price our energy industrial product and measure our shipments in square feet. We believe the square foot operating metric allows us and our investors to measure our manufacturing capacity and energy industrial product shipments on a uniform and consistent basis. The following chart sets forth energy industrial product shipments in square feet associated with recognized revenue, including revenue recognized over time utilizing the input method, for the periods presented:

	Three Months Ended March 31,	
	2023	2022
	(In thousands)	
Product shipments in square feet	8,183	8,163

Adjusted EBITDA

We use Adjusted EBITDA, a non-GAAP financial measure, as a means to assess our operating performance. We define Adjusted EBITDA as net income (loss) before interest expense, taxes, depreciation, amortization, stock-based compensation expense and other items, from time to time, which we do not believe are indicative of our core operating performance. Adjusted EBITDA is a supplemental measure of our performance that is not presented in accordance with U.S. GAAP. Adjusted EBITDA should not be considered as an alternative to net income (loss) or any other measure of financial performance calculated and presented in accordance with U.S. GAAP. In addition, our definition and presentation of Adjusted EBITDA may not be comparable to similarly titled measures presented by other companies.

We use Adjusted EBITDA:

- as a measure of operating performance because it does not include the impact of items that we do not consider indicative of our core operating performance;
- for planning purposes, including the preparation of our annual operating budget;
- to allocate resources to enhance the financial performance of our business; and
- as a performance measure used under our bonus plan.

We also believe that the presentation of Adjusted EBITDA provides useful information to investors with respect to our results of operations and in assessing the performance and value of our business. Various measures of EBITDA are widely used by investors to measure a company's operating performance without regard to items that can vary substantially from company to company depending upon financing and accounting methods, book values of assets, capital structures and the methods by which assets were acquired.

Although measures similar to Adjusted EBITDA are frequently used by investors and securities analysts in their evaluation of companies, we understand that Adjusted EBITDA has limitations as an analytical tool, and you should not consider it in isolation or as a substitute for net income (loss), income (loss) from operations, net cash provided by (used in) operating activities or an analysis of our results of operations as reported under U.S. GAAP. Some of these limitations are:

- Adjusted EBITDA does not reflect our historical cash expenditures or future requirements for capital expenditures or other contractual commitments;
- Adjusted EBITDA does not reflect changes in, or cash requirements for, our working capital needs;

- Adjusted EBITDA does not reflect stock-based compensation expense;
- Adjusted EBITDA does not reflect our income tax expense or cash requirements to pay our income taxes;
- Adjusted EBITDA does not reflect our interest expense, or the cash requirements necessary to service interest or principal payments on our debt;
- although depreciation, amortization and impairment charges are non-cash charges, the assets being depreciated, amortized or impaired will often have to be replaced in the future, and Adjusted EBITDA does not reflect any cash requirements for these replacements; and
- other companies in our industry may calculate EBITDA or Adjusted EBITDA differently than we do, limiting their usefulness as a comparative measure.

Because of these limitations, our Adjusted EBITDA should not be considered as a measure of discretionary cash available to us to reinvest in the growth of our business or as a measure of cash available for us to meet our obligations.

To properly and prudently evaluate our business, we encourage you to review the U.S. GAAP financial statements included elsewhere in this Quarterly Report on Form 10-Q, and not to rely on any single financial measure to evaluate our business.

The following table presents a reconciliation of net loss, the most directly comparable U.S. GAAP measure, to Adjusted EBITDA for the periods presented:

	Three Months Ended March 31,	
	2023	2022
	(In thousands)	
Net loss	\$ (16,796)	\$ (19,484)
Depreciation and amortization	2,704	2,129
Stock-based compensation ⁽¹⁾	2,267	1,828
Interest (income) expense	(2,112)	860
Adjusted EBITDA	\$ (13,937)	\$ (14,667)

⁽¹⁾ Represents non-cash stock-based compensation related to vesting and modifications of stock option grants, vesting of restricted stock units and vesting of restricted common stock.

Our financial performance, including such measures as net income (loss), earnings per share and Adjusted EBITDA, are affected by a number of factors including volume and mix of aerogel products sold, average selling prices, our material costs and manufacturing expenses, the costs associated with capacity expansions and start-up of additional production capacity, and the amount and timing of operating expenses. Accordingly, we expect that our net income (loss), earnings per share and Adjusted EBITDA will vary from period to period.

We expect to maintain strong revenue growth during 2023 driven by a continued post-COVID recovery in the energy industrial market, accelerating demand in the electric vehicle market and continued market share gains in the sustainable insulation materials market. Our expectation to maintain strong revenue growth is based, in part, on our OEM customers' production volume forecasts and targets as well as our expectation to successfully scale our manufacturing capabilities and address any potential supply chain issues to meet this expected demand. As a result, we expect to experience a decrease in both net loss and Adjusted EBITDA during 2023.

Components of Our Results of Operations

Revenue

We recognize revenue from the sale of our energy industrial aerogel products and thermal barriers. Revenue is recognized upon the satisfaction of contractual performance obligations.

We record deferred revenue for sales when (i) we have delivered products, but other revenue recognition criteria have not been satisfied, or (ii) payments have been received in advance of the completion of required performance obligations.

We project revenue growth during 2023 due to accelerating demand in the electric vehicle market and continued market share gains in the sustainable insulation materials market.

Cost of Revenue

Cost of product revenue consists primarily of materials and manufacturing expense. Cost of product revenue is recorded when the related product revenue is recognized.

Material is our most significant component of cost of product revenue and includes fibrous batting, silica materials and additives. Material costs as a percentage of product revenue vary from product to product due to differences in average selling prices, material requirements, product thicknesses, and manufacturing yields. In addition, we provide warranties for our products and record the estimated cost within cost of revenue in the period that the related revenue is recorded or when we become aware that a potential warranty claim is probable and can be reasonably estimated. As a result of these factors, material costs as a percentage of product revenue will vary from period to period due to changes in the mix of aerogel products sold, the costs of our raw materials or the estimated cost of warranties. In addition, global supply chain disturbances, increased reliance on foreign materials procurement, industrial gas supply constraints, increases in the cost of our raw materials, and other factors may significantly impact our material costs and have a material impact on our operations. We expect that material costs will increase in absolute dollars during 2023 due to projected growth in product shipments, but decrease as a percentage of revenue due to projected increases in average selling prices, improved manufacturing, and fabrication yields and a favorable mix of products sold.

Manufacturing expense is also a significant component of cost of revenue. Manufacturing expense includes labor, utilities, maintenance expense, and depreciation on manufacturing assets. Manufacturing expense also includes stock-based compensation of manufacturing employees and shipping costs. We expect that manufacturing expense will increase in absolute dollars and decrease as a percentage of revenue during 2023 due to increased staffing and spending levels in support of our thermal barrier business, including the start-up and operation of an automated fabrication facility in Monterrey, Mexico. We are also continuing to monitor the impact of engaging one or more contract manufacturers in China to supply our aerogel products for the energy industrial market beginning in 2024 on our manufacturing expense and cost of product revenue.

In total, we expect that cost of product revenue will increase in absolute dollars during 2023 versus 2022 and decrease as a percentage of revenue versus 2022 driven by the costs to support our expected higher run-rate revenue in future periods.

Gross Profit

Our gross profit as a percentage of revenue is affected by a number of factors, including the volume of products produced and sold, the mix of products sold, average selling prices, our material and manufacturing costs, realized capacity utilization and the costs associated with expansions and start-up of production capacity. Accordingly, we expect our gross profit to vary significantly in absolute dollars and as a percentage of revenue from period to period.

During 2023, we expect gross profit to increase in both absolute dollars and as a percentage of total revenue due to the combination of a projected increase in total revenue combined with projected reduction in material costs as a percentage of total revenue, offset, in part, by a projected increase in manufacturing expense as a percentage of revenue.

In the longer term, we expect gross profit to improve in absolute dollars and as a percentage of revenue due to expected increases in total revenue, production volumes and manufacturing productivity. In addition, we expect the gross profit improvement derived from the increases in revenue, volume and productivity will be supported by the continued implementation of lower cost product formulations and realization of material purchasing efficiencies.

Operating Expenses

Operating expenses consist of research and development, sales and marketing, and general and administrative expenses. Operating expenses include personnel costs, legal fees, professional fees, service fees, insurance premiums, travel expense, facilities related costs and other costs, expenses and fees. The largest component of our operating expenses is personnel costs, consisting of salaries, benefits, incentive compensation and stock-based compensation. In any particular period, the timing and extent of personnel additions or reductions, legal activities, including patent enforcement actions, marketing programs, research efforts and a range of similar activities or actions could materially affect our operating expenses, both in absolute dollars and as a percentage of revenue.

During 2023, we expect to continue to hire additional personnel and incur additional operating expenses to support the anticipated multi-year growth in our PyroThin thermal barrier business. As a result, we expect that operating expenses will increase in absolute dollars, and remain consistent as a percentage of revenue during the year. In the longer term, we expect that operating expenses will increase in absolute dollars, but decrease as a percentage of revenue.

Research and Development Expenses

Research and development expenses consist primarily of expenses for personnel engaged in the development of next generation aerogel compositions, form factors and manufacturing technologies. These expenses also include testing services, prototype expenses, consulting services, trial formulations for new products, equipment depreciation, facilities costs and related overhead. We expense research and development costs as incurred. We expect to continue to devote substantial resources to the development of new aerogel technologies, including our carbon aerogel battery materials. We believe that these investments are necessary to maintain and improve our competitive position. We also expect to continue to invest in research and engineering personnel and the infrastructure required in support of their efforts. While we expect our research and development expenses will increase in absolute dollars but decrease as a percentage of revenue in the longer term, in 2023 we expect these expenses will increase in both absolute dollars and as a percentage of revenue.

Sales and Marketing Expenses

Sales and marketing expenses consist primarily of personnel costs, incentive compensation, marketing programs, travel and related costs, consulting expenses and facilities related costs. We expect that sales and marketing expenses will increase in absolute dollars and remain consistent as a percentage of revenue during 2023 principally due to an increase in compensation associated with the addition of personnel in support of our PyroThin thermal barrier business. In the longer term, we expect that sales and marketing expenses will increase in absolute dollars but decrease as a percentage of revenue.

General and Administrative Expenses

General and administrative expenses consist primarily of personnel costs, legal expenses, consulting and professional services, audit fees, compliance with securities, corporate governance and related laws and regulations, investor relations expenses and insurance premiums, including director and officer insurance.

We expect our general and administrative expenses to increase as we add general and administrative personnel to support the anticipated growth of our business. We also expect that the patent enforcement actions, described in more detail under “Legal Proceedings” in Part I, Item 3 of our Annual Report on Form 10-K for the year ended December 31, 2022 and “Legal Proceedings” in Part II, Item 1 of this Quarterly Report on Form 10-Q, if protracted, could result in significant legal expense over the medium to long-term. We expect that our general and administrative expenses will increase in absolute dollars but decrease as a percentage of revenue in the longer term. In 2023, we expect such expenses will increase in both absolute dollars and as a percentage of revenue.

Interest Income (Expense), Convertible Note - Related Party

Interest expense, convertible note - related party is net of the capitalized interest related to the \$100.0 million in aggregate principal amount of our Convertible Senior PIK Toggle Notes.

Interest Income (Expense), Net

Interest expense, net consists of interest expense related to our revolving credit facility and included interest earned on the cash balances invested in deposit accounts, money market accounts, and high-quality debt securities issued by the U.S. government. We terminated our revolving credit facility agreement on November 28, 2022.

Provision for Income Taxes

We have incurred net losses since inception and have not recorded benefit provisions for U.S. federal income taxes or state income taxes since the tax benefits of our net losses have been offset by valuation allowances due to the uncertainty associated with the utilization of net operating loss carryforwards.

Results of Operations

Three months ended March 31, 2023 compared to the three months ended March 31, 2022

The following tables set forth a comparison of the components of our results of operations for the periods presented:

Revenue

	Three Months Ended March 31,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
Revenue:						
Energy industrial	\$ 33,875	74 %	\$ 30,775	80 %	\$ 3,100	10 %
Thermal barrier	11,711	26 %	7,632	20 %	4,079	53 %
Total revenue	\$ 45,586	100 %	\$ 38,407	100 %	\$ 7,179	19 %

Total revenue increased \$7.2 million, or 19%, to \$45.6 million for the three months ended March 31, 2023 from \$38.4 million in the comparable period in 2022. The increase in total revenue was the result of an increase in both thermal barrier and energy industrial revenue.

The following chart sets forth energy industrial product shipments in square feet associated with recognized revenue, including revenue recognized over time utilizing the input method, for the periods presented:

	Three Months Ended March 31,		Change	
	2023	2022	Amount	Percentage
Product shipments in square feet (in thousands)	8,183	8,163	20	0 %

Energy industrial revenue increased by \$3.1 million, or 10%, to \$33.9 million for the three months ended March 31, 2023 from \$30.8 million in the comparable period in 2022. This increase was driven by a more favorable mix of product shipments in the global petrochemical and refinery markets in Asia, project-based demand in the subsea market, offset, in part, by a decrease in the volume of shipments in the global petrochemical and refinery markets of North America and Europe.

Energy industrial revenue for the three months ended March 31, 2023 included \$10.9 million to a North American distributor. Energy industrial revenue for the three months ended March 31, 2022 included \$11.0 million to a North American distributor.

The average selling price per square foot of our energy industrial products increased by \$0.37, or 10%, to \$4.14 per square foot for the three months ended March 31, 2023 from \$3.77 per square foot for the three months ended March 31, 2022. The increase in average selling price principally reflected the impact of a change in the mix of products sold. This increase in average selling price had the effect of increasing product revenue by \$3.0 million for the three months ended March 31, 2023 from the comparable period in 2022.

In volume terms, energy industrial product shipments remained consistent at 8.2 million square feet for both of the three months ended March 31, 2023 and 2022.

Thermal barrier revenue was \$11.7 million for the three months ended March 31, 2023 as compared to \$7.6 million for the three months ended March 31, 2022. During the three months ended March 31, 2023 and 2022, thermal barrier revenue included \$9.7 million and \$6.1 million to a major U.S. automotive OEM, respectively.

Cost of Revenue

	Three Months Ended March 31,						Change	
	2023			2022				
	Amount	Percentage of Related Revenue	Percentage of Total Revenue	Amount	Percentage of Related Revenue	Percentage of Total Revenue	Amount	Percentage
(\$ in thousands)								
Cost of revenue:								
Energy industrial	\$ 24,994	74 %	55 %	\$ 27,778	90 %	72 %	\$ (2,784)	(10)%
Thermal barrier	15,506	132 %	34 %	12,417	163 %	32 %	3,089	25 %
Total cost of revenue	<u>\$ 40,500</u>	89 %	89 %	<u>\$ 40,195</u>	105 %	105 %	<u>\$ 305</u>	1 %

Total cost of revenue increased \$0.3 million, or 1%, to \$40.5 million for the three months ended March 31, 2023 from \$40.2 in the comparable period in 2022. The increase in total cost of revenue was the result of increases in thermal barrier cost of revenue, offset by a decrease in energy industrial cost of revenue.

Energy industrial cost of revenue decreased \$2.8 million, or 10%, to \$25.0 million for the three months ended March 31, 2023 from \$27.8 million in the comparable period in 2022. The \$2.8 million decrease was the result of a \$1.6 million decrease in manufacturing and other operating costs and a \$1.2 million decrease in material costs due to change in the product mix from the comparable period in 2022.

Thermal barrier cost of revenue increased \$3.1 million to \$15.5 million for the three months ended March 31, 2023 as compared to \$12.4 million for the three months ended March 31, 2022. The \$3.1 million increase was the result of a \$4.2 million increase in manufacturing costs, offset by a \$1.1 million decrease in material costs. The increase in manufacturing costs was driven by increases in depreciation and facility costs of \$2.6 million and other manufacturing and operating costs of \$1.6 million.

Gross Profit

	Three Months Ended March 31,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
(\$ in thousands)						
Gross profit:						
Energy industrial	\$ 8,881	26 %	\$ 2,997	10 %	\$ 5,884	196 %
Thermal barrier	(3,795)	(32)%	(4,785)	(63)%	990	21 %
Total gross (loss) profit	<u>\$ 5,086</u>	11 %	<u>\$ (1,788)</u>	(5)%	<u>\$ 6,874</u>	384 %

Gross profit increased by \$6.9 million, or 384%, to \$5.1 million for the three months ended March 31, 2023 from \$(1.8) million in the comparable period in 2022. The increase in gross profit was the result of the \$7.2 million increase in total revenue, offset, in part, by the \$0.3 million increase in total cost of revenue. The increase in gross profit reflects the decrease in costs and additional resources to support our expected higher run-rate revenue in future periods for both our energy industrial and thermal barrier products from the comparable period in 2022.

Research and Development Expenses

	Three Months Ended March 31,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
(\$ in thousands)						
Research and development expenses	\$ 4,099	9 %	\$ 3,592	9 %	\$ 507	14 %

Research and development expenses increased by \$0.5 million, or 14%, to \$4.1 million for the three months ended March 31, 2023 from \$3.6 million in the comparable period in 2022. The \$0.5 million increase reflects an increase in depreciation expenses of \$0.3 million and compensation and related costs of \$0.2 million.

Research and development expenses as a percentage of total revenue were 9% for both the three months ended March 31, 2023 and 2022.

Sales and Marketing Expenses

	Three Months Ended March 31,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Sales and marketing expenses	\$ 7,713	17 %	\$ 6,018	16 %	\$ 1,695	28 %

Sales and marketing expenses increased by \$1.7 million, or 28%, to \$7.7 million for the three months ended March 31, 2023 from \$6.0 million in the comparable period in 2022. The \$1.7 million increase was principally the result of increases in compensation and related costs of \$1.0 million, facility related expenditures of \$0.4 million and other sales and marketing expenses of \$0.3 million.

Sales and marketing expenses as a percentage of total revenue increased to 17% for the three months ended March 31, 2023 from 16% in the comparable period in 2022, due principally to the increase in compensation and related expenses associated with an increase in sales and business development personnel.

General and Administrative Expenses

	Three Months Ended March 31,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
General and administrative expenses	\$ 12,182	27 %	\$ 7,226	19 %	\$ 4,956	69 %

General and administrative expenses increased by \$5.0 million, or 69%, to \$12.2 million for the three months ended March 31, 2023 from \$7.2 million in the comparable period in 2022. The \$5.0 million increase was the result of increases in compensation and related costs of \$3.8 million and professional services expenses of \$1.2 million.

General and administrative expenses as a percentage of total revenue increased to 27% for the three months ended March 31, 2023 from 19% in the comparable period in 2022.

Interest Income (Expense), net

	Three Months Ended March 31,				Change	
	2023		2022			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Interest income (expense):						
Interest (expense), related party	\$ (275)	(1) %	\$ (819)	(2) %	\$ 544	(66) %
Interest income (expense), net	2,387	5 %	(41)	—	2,428	NM
Total interest income (expense), net	\$ 2,112	5 %	\$ (860)	0 %	\$ 2,972	(346) %

Interest income (expense), net increased by \$3.0 million to \$2.1 million of interest income for the three months ended March 31, 2023 from \$0.9 million of interest expense in the comparable period in 2022. The \$3.0 million increase was the result of \$2.4 million of interest income and a \$0.6 million net impact of capitalized interest relating to our Convertible Note in the comparable period in 2022.

Liquidity and Capital Resources

Overview

We have experienced significant losses and invested substantial resources since our inception to develop, commercialize and protect our aerogel technology and to build a manufacturing infrastructure capable of supplying aerogel products at the volumes and costs required by our customers. These investments have included research and development and other operating expenses, capital expenditures and investment in working capital balances.

Our long-term financial projections anticipate revenue growth, increasing levels of gross profit, and improved cash flows from operations. To meet expected growth in demand for our aerogel products in the electric vehicle market, we have been in the process of

expanding our aerogel blanket capacity by constructing a second manufacturing plant in Bulloch County, Georgia. However, in order to manage the development of the second plant so that its increased capacity comes online in a manner that aligns with our current expectations as to demand from our EV customers, we are extending the timeframe for construction and commissioning of the second plant until such time as its capacity is supported by increased demand. In the meantime, and until we restart construction, we expect to be able to substantially reduce our planned capital expenditures for 2023 and 2024. At the same time, we believe that productivity improvements in our existing Rhode Island facility combined with supply of our energy industrial products from one or more contract manufacturers in China beginning in 2024 will permit us to achieve a target revenue capacity of approximately \$550.0 million in 2024 and prior to the completion and start-up of the second plant. Nonetheless, there can be no assurance as to when we will restart construction on the second plant and we are in the process of determining what remaining or additional capital expenditures and other costs may result from the decision to extend the timeframe for construction of the second plant. There can also be no assurance that our contract manufacturing strategy of meeting the demand of our energy industrial customers with supply from one or more contract manufacturers in China will provide us with adequate manufacturing capacity or supply for that expected demand. Furthermore, if and when we restart construction on the second plant, further cost inflation and/or supply chain disruptions, as well as potential changes in the scope of the facilities, could lead to increases to our prior estimates for completion of the second plant.

We are also increasing our investment in the research and development of next-generation aerogel products and technologies. During 2023, we will continue to develop aerogel products and technologies for the electric vehicle market. We believe the commercial potential for our technology in the electric vehicle market is significant. To meet the anticipated revenue growth and take advantage of this market opportunity, we are adding personnel, incurring additional operating expenses, and planning to construct a carbon aerogel battery materials facility, among other items.

In February 2022, we sold and issued to an affiliate of Koch \$100.0 million in aggregate principal amount of our Convertible Senior PIK Toggle Notes. In addition, in March 2022, pursuant to a securities purchase agreement dated February 15, 2022, we sold to an affiliate of Koch 1,791,986 shares of our common stock, at a price of \$27.902 per share, for net proceeds of \$49.9 million after deducting fees and offering expenses of \$0.1 million.

We believe that our March 31, 2023 cash and cash equivalents balance of \$207.5 million will be sufficient to support current operating requirements, current research and development activities and the initial capital expenditures required to support the evolving commercial opportunities in the electric vehicle market and other strategic business opportunities.

However, we plan to supplement our cash balance and available credit with equity financings, debt financings, customer prepayments or technology licensing fees to provide the additional capital necessary to purchase the capital equipment, construct the new facilities and complete the aerogel capacity expansions required to support our evolving commercial opportunities and strategic business initiatives. We also intend to enter into a new revolving credit facility. We believe that the consummation of equity financings could potentially result in an ownership change under Section 382 of the Internal Revenue Code. Such an ownership change would lead to the use of our net operating loss carryforwards being restricted. Our inability to use a substantial portion of our net operating loss carryforwards would result in a higher effective tax rate and adversely affect our financial condition and results of operations.

Primary Sources of Liquidity

Our principal sources of liquidity are currently our cash and cash equivalents. Cash and cash equivalents consist primarily of cash, money market accounts, and sweep accounts on deposit with banks. As of March 31, 2023, we had \$207.5 million of unrestricted cash and cash equivalents.

In February 2022, we sold and issued to an affiliate of Koch \$100.0 million in aggregate principal amount of our Convertible Senior PIK Toggle Notes. In addition, in March 2022, pursuant to a securities purchase agreement dated February 15, 2022, we sold to an affiliate of Koch 1,791,986 shares of our common stock, at a price of \$27.902 per share, for net proceeds of \$49.9 million after deducting fees and offering expenses of \$0.1 million.

On March 16, 2022, we entered into a sales agreement for an ATM offering program with Cowen and Company, LLC and Piper Sandler & Co., as our sales agents. During the year ended December 31, 2022, we sold 5,241,400 shares of our common stock through the 2022 ATM offering program and received net proceeds of \$72.7 million.

On November 28, 2022, our wholly owned subsidiary, Aspen Aerogels Georgia, LLC, entered into a \$100.0 million loan agreement with GM for which the proceeds to be used for the construction and operation of our planned aerogel manufacturing facility in Bulloch County, Georgia. The agreement allows for borrowings beginning in 2023.

On November 29, 2022, we completed an underwritten public offering of 29,052,631 shares of our common stock at a public offering price of \$9.50 per share. We received net proceeds of \$267.5 million after deducting underwriting discounts and commissions of \$8.1 million and offering expenses of approximately \$0.5 million.

Analysis of Cash Flow

Net Cash Used in Operating Activities

During the three months ended March 31, 2023, we used \$24.7 million in net cash in operating activities, as compared to the use of \$22.8 million in net cash during the comparable period in 2022, an increase in the use of cash of \$1.9 million. This increase in use of cash was the result of cash provided by net loss adjusted for non-cash items of \$3.3 million and in net cash used by changes in operating assets and liabilities of \$5.2 million.

During the three months ended March 31, 2022, we used \$22.8 million in net cash in operating activities, as compared to the use of \$1.9 million in net cash during the comparable period in 2021, an increase in the use of cash of \$20.9 million. This increase in use of cash was the result of increases in net loss adjusted for non-cash items of \$11.7 million and in net cash used by changes in operating assets and liabilities of \$9.2 million.

Net Cash Used in Investing Activities

Net cash used in investing activities is for capital expenditures for machinery and equipment principally to improve the throughput, efficiency and capacity of our East Providence facility and engineering designs and construction costs for the planned aerogel manufacturing facility in Bulloch County, Georgia. Net cash used in investing activities for the three months ended March 31, 2023 and 2022 was \$49.4 million and \$14.5 million, respectively.

Net Cash Provided by Financing Activities

Net cash used in financing activities for the three months ended March 31, 2023 totaled \$0.4 million and consisted of \$0.4 million in cash used for payments made for employee tax withholdings associated with the vesting of restricted stock units, offset, in part, by less than \$0.1 million in proceeds from employee stock option exercises.

Net cash provided by financing activities for the three months ended March 31, 2022 totaled \$165.9 million and consisted of \$99.8 million in net proceeds from the issuance of convertible debt, \$49.9 million in net proceeds from the private placement of our common stock, \$23.3 million in net proceeds from the ATM offering program, and less than \$0.1 million in proceeds from employee stock option exercises, offset, in part, by \$4.7 million in cash used for payments made for repayments of a prepayment liability and \$2.4 million in cash used for payments made for employee tax withholdings associated with the vesting of restricted stock units.

Contractual Obligations and Commitments

There have been no material changes to our contractual obligations and commitments as reported in our Annual Report.

Recent Accounting Pronouncements

Information regarding new accounting pronouncements is included in note 2 to our unaudited consolidated financial statements contained in Item 1 of this Quarterly Report on Form 10-Q.

Critical Accounting Policies and Estimates

Our financial statements are prepared in accordance with U.S. GAAP. The preparation of our financial statements and related disclosures requires us to make estimates, assumptions and judgments that affect the reported amount of assets, liabilities, revenue, costs and expenses and related disclosures. We believe that the estimates, assumptions and judgments involved in these accounting policies have the greatest potential impact on our financial statements and, therefore, we consider these to be our critical accounting policies. Accordingly, we evaluate our estimates and assumptions on an ongoing basis. Our actual results may differ from these estimates under different assumptions and conditions. See our Annual Report and note 2 to our consolidated financial statements

included elsewhere in this Quarterly Report on Form 10-Q for information about these critical accounting policies, as well as a description of our other significant accounting policies.

Certain Factors That May Affect Future Results of Operations

The SEC encourages companies to disclose forward-looking information so that investors can better understand a company's future prospects and make informed investment decisions. This Quarterly Report on Form 10-Q contains such "forward-looking statements" within the meaning of the Private Securities Litigation Reform Act of 1995. These statements involve known and unknown risks, uncertainties and other important factors, which may cause our actual results, performance or achievements to be materially different from any future results, performances or achievements expressed or implied by the forward-looking statements. Forward-looking statements include, but are not limited to, statements about: the expected future growth of the market for our aerogel products and our continued gain in market share, in particular in the electric vehicle market, the energy infrastructure insulation market, the lithium-ion battery thermal barrier markets, and other markets we target; our beliefs in the appropriateness of our assumptions, the accuracy of our estimates regarding expenses, loss contingencies, future revenues, revenue capacity, future profits, uses of cash, available credit, capital requirements, and the need for additional financing to operate our business, including to complete the planned construction and development of our second manufacturing facility in Bulloch County, Georgia, or fabrication operations in Monterrey, Mexico, and to fund our planned strategic business initiatives; the performance of our aerogel blankets; our expectation that we will be successful in obtaining, enforcing and defending our patents against competitors and that such patents are valid and enforceable; our belief that our products possess strong competitive advantages over traditional insulation materials, including the superior thermal performance and the thin, easy-to-use and durable blanket form of our products; our expectations regarding the investment to open a second manufacturing facility in Georgia, the extended construction and commissioning timeframe for the planned second manufacturing facility, our efforts to manage the construction of the second plant to align with our expectations of demand from EV customers, and our ability to complete the second plant; the anticipated capacity expansion as a result of the planned second manufacturing facility in Georgia, if it were to be completed; our estimates of annual production capacity; beliefs about the commercial potential for our technology in the electric vehicle market; beliefs about our ability to produce and deliver products to electric vehicle customers; beliefs about Aspen's contracts with the major U.S. automotive manufacturer; our expectations about the size and timing of awarded business in the electric vehicle market, future revenues and profit margins, arising from our supply relationship and contract with automotive OEMs and our ability to win more business and increase revenue in the electric vehicle market; beliefs about the performance of our thermal barrier products in the battery systems of electric vehicles; the current or future trends in the energy, energy infrastructure, chemical and refinery, LNG, sustainable building materials, electric vehicle thermal barrier, electric vehicle battery materials or other markets and the impact of these trends on our business; our investments in the electric vehicle market and aerogel technology platform; our beliefs about the financial metrics that are indicative of our core performance; our expectations about the effect of manufacturing capacity on financial metrics such as Adjusted EBITDA; our expectations about future revenues, expenses, gross profit, net loss, loss per share and Adjusted EBITDA, sources and uses of cash, capital requirements and the sufficiency of our existing cash balance and available credit; our beliefs about the outcome, effects or estimated costs of current or potential litigation or their respective timing, including expected legal expense in connection with our patent enforcement actions; our expectations about future material costs and manufacturing expenses as a percentage of revenue, including the impact of engaging one or more contract manufacturers in China for supply of our energy industrial products; our expectation about the ability of the Chinese contract manufacturers that we engage to consistently supply the aerogel product that we order in a timely manner; our expectations of future gross profit and the effect of manufacturing expenses, manufacturing capacity and productivity on gross profit; our expectations about our resources and other investments in new technology and related research and development activities and associated expenses; our expectations about short and long term (a) research and development (b) general and administrative and (c) sales and marketing expenses; our expectations of revenue growth, increased gross profit, and improving cash flows over the long term; our intentions about managing capital expenditures and working capital balances; our expectations about potential sources of future financing; and our statements about the impact of major public health concerns, including the COVID-19 pandemic or other pandemics arising globally, and the future, and currently unknown extent of, the impact of the COVID-19 pandemic on our business and operations.

Words such as "may," "will," "anticipate," "estimate," "expects," "projects," "intends," "plans," "believes" and words and terms of similar substance used in connection with any discussion of future operating or financial performance, identify forward-looking statements. All forward-looking statements are management's present expectations of future events and are subject to a number of risks and uncertainties that could cause actual results to differ materially and adversely from those described in the forward-looking statements. These risks include, but are not limited to, those set forth in this Quarterly Report on Form 10-Q and under the heading "Risk Factors" contained in Item 1A of our Annual Report.

In light of these assumptions, risks and uncertainties, the results and events discussed in the forward-looking statements contained in this Quarterly Report on Form 10-Q might not occur. Stockholders and other readers are cautioned not to place undue reliance on the forward-looking statements, which speak only as of the date of this Quarterly Report on Form 10-Q. We are not under

any obligation, and we expressly disclaim any obligation, to update or alter any forward-looking statements, whether as a result of new information, future events or otherwise. All subsequent forward-looking statements attributable to Aspen Aerogels, Inc. or to any person acting on its behalf are expressly qualified in their entirety by the cautionary statements contained or referred to in this section.

Item 3. Quantitative and Qualitative Disclosures About Market Risk.

Market risk represents the risk of loss that may impact our financial position due to adverse changes in financial market prices and rates. Our market risk exposure results primarily from fluctuations in interest rates, as well as from inflation. In the normal course of business, we are exposed to market risks, including changes in interest rates which affect our cash flows. We may also face additional exchange rate risk in the future as we expand our business internationally.

Interest Rate Risk

We are exposed to changes in interest rates in the normal course of our business. As of March 31, 2023, we had unrestricted cash and cash equivalents of \$207.5 million. These amounts were held for working capital and capital expansion purposes and were invested primarily in deposit accounts, money market accounts, and high-quality debt securities issued by the U.S. government via cash sweep accounts at major financial institutions in North America. Due to the short-term nature of these investments, we believe that our exposure to changes in the fair value of our cash as a result of changes in interest rates is not material.

As of March 31, 2023, we had a convertible note outstanding with principal balance of \$107.8 million. Our convertible note bears interest at the Secured Overnight Financing Rate (SOFR) plus 5.50% per annum if interest is paid in cash, or, if interest is paid in-kind as an increase in the principal amount of the outstanding note, at the SOFR plus 6.50% per annum. Under the terms of the investment, SOFR has a floor of 1% and a cap of 3%. Interest is paid semi-annually in arrears on June 30 and December 30. We, at our option, are permitted to settle each semi-annual interest payment in cash, in-kind, or any combination thereof.

As of March 31, 2023, we had \$0.6 million of restricted cash to support our outstanding letters of credit to secure obligations under certain commercial contracts and other obligations. We terminated our revolving credit facility agreement on November 28, 2022.

Inflation Risk

Although we expect that our operating results will be influenced by general economic conditions, we do not believe that inflation has had a material effect on our results of operations during the periods presented in this report. However, our business may be affected by inflation in the future.

Foreign Currency Exchange Risk

We are subject to inherent risks attributed to operating in a global economy. Principally all our revenue, receivables, purchases and debts are denominated in U.S. dollars.

Item 4. Controls and Procedures.

(a) Evaluation of Disclosure Controls and Procedures.

We maintain disclosure controls and procedures that are designed to ensure that information required to be disclosed in the reports that we file or submit under the Securities Exchange Act of 1934, as amended (the Exchange Act), is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms and is accumulated and communicated to our management, including our principal executive officer and principal financial officer, or persons performing similar functions, as appropriate to allow timely decisions regarding required disclosure.

As of March 31, 2023, our management, with the participation of our principal executive officer and principal financial officer, evaluated the effectiveness of our disclosure controls and procedures (as defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act). Our management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving their objectives, and management necessarily applies its judgment in evaluating the cost-benefit relationship of possible controls and procedures. Based on such evaluation, our principal executive officer and principal financial officer have concluded that, as of March 31, 2023, our disclosure controls and procedures were effective to ensure that information required to be disclosed by us in the reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms, and is accumulated and communicated to our management, including our principal executive officer and principal financial officer, or persons performing similar functions, as appropriate to allow timely decisions regarding required disclosure.

(b) *Changes in Internal Controls.*

During the three months ended March 31, 2023, there were no changes in our internal control over financial reporting, as such term is defined in Rules 13a-15(f) and 15(d)-15(f) promulgated under the Exchange Act, that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

PART II — OTHER INFORMATION

Item 1. Legal Proceedings.

We are involved in various legal claims and proceedings in the normal course of operations. We believe the outcome of these matters will not have a material adverse effect on our consolidated financial position, results of operations or liquidity, except as described in Part 1, Item 3. “Legal Proceedings” of our Annual Report on Form 10-K. Since the filing of our Form 10-K, there have been no material changes in our legal proceedings from those disclosed therein, other than as noted below.

On April 18, 2023, we filed a patent infringement complaint at the Seoul Central District Court and a petition for investigation of unfair trade practices in the Korea Trade Commission (KTC). The complaint and petition allege that Beerenberg Services AS, Beerenberg Korea Ltd., Bronx Holdings Pte. Ltd., and Bronx (China) Co., Ltd. (collectively, “Beerenberg”) have infringed several of our Korean patents in connection with the import and sale of certain aerogel products in Korea. The asserted patents include (a) Korean patents related to high performance reinforced aerogel compositions and (b) Korean counterparts of the patents previously successfully asserted against Nano Tech Co., Ltd. (“Nano”) and Guangdong Alison Hi Tech., Ltd. (“Alison”) in Germany and the US. We are seeking injunctive relief and monetary damages against the defendants.

Item 1A. Risk Factors.

The ownership of our common stock involves a number of risks and uncertainties. When evaluating the Company and our business before making an investment decision regarding our securities, potential investors should carefully consider the risk factors and uncertainties described in Part 1, Item 1A. “Risk Factors” of our Annual Report on Form 10-K. Since the filing of our Form 10-K, there have been no material changes in our risk factors from those disclosed therein, other than as provided below.

We are engaging third-party contract manufacturers in China to supply our energy industrial products beginning in 2024. If such contract manufacturers are unable to manufacture and deliver a sufficient quantity of high-quality products on a timely and cost-efficient basis, our net revenue and business operations may be harmed and our reputation may suffer.

We are engaging one or more contract manufacturers in China for supply of our energy industrial products beginning in 2024, which we believe will enable us to achieve a target revenue capacity of approximately \$550.0 million in 2024 and prior to the completion and start-up of our planned second manufacturing plant. If our contract manufacturers are unable to deliver the required aerogel product on a timely basis, we may experience delays in delivering our finished aerogel product to customers in the energy industrial market. In addition, because our third party contract manufacturers have manufacturing facilities in China, their ability to provide us with adequate supplies of high-quality products on a timely and cost-efficient basis is subject to a number of additional risks and uncertainties, including political, social and economic instability and other factors that could impact the shipment of supplies. If our manufacturers are unable to provide us with adequate supplies of high-quality aerogel products on a timely and cost-efficient basis, our operations could be disrupted and our revenue and business operations may suffer. Moreover, if our third-party contract manufacturers cannot consistently produce high-quality products that are free of defects, we may experience a loss of customers, which may also reduce our revenues and may harm our reputation and brand. Furthermore, our third-party contract manufacturers may become subject to various supply chain disruptions, including but not limited as a result of COVID-19, other pandemics or public health crises, and geopolitical disputes and conflicts, any of which could slow or halt the delivery of products to us and increase the price of certain materials due to resulting increases in costs of raw materials and shipping costs.

Our potential inability to adequately protect our intellectual property as a result of engaging contract manufacturers in China for the supply of our aerogel products for our customers in the energy industrial market could negatively impact our performance.

In connection with our engagement of contract manufacturers in China, we expect to implement customary manufacturer safeguards onsite, such as the use of confidentiality agreements with employees, to protect our proprietary information and technologies during the manufacturing process of our aerogel products for the energy industrial market. However, these safeguards may not effectively prevent unauthorized use of such information and technical know-how, or prevent the contract manufacturer from retaining them. Although the courts in China are increasing and broadening their protection of intellectual property rights, the legal regime governing intellectual property rights in China is relatively immature and it is often difficult to create and enforce intellectual property rights or protect trade secrets there. We face risks that our proprietary information may not be afforded the same protection in China as it is in countries with well-developed intellectual property laws, and local laws may not provide an adequate remedy in the event of unauthorized disclosure of confidential information. Costly and time-consuming litigation could be necessary to enforce and determine the scope of our proprietary rights in China, and failure to obtain or maintain trade secret protection could adversely affect our competitive business position. In the event that the third-party contract manufacturer of our proprietary aerogel product misappropriates our intellectual property, our business, prospects and financial condition could be materially and adversely affected.

Item 2. Unregistered Sales of Equity Securities and Use of Proceeds.

(a) Unregistered Sales of Equity Securities.

None.

(b) Use of Proceeds from Initial Public Offering of Common Stock.

Not applicable.

(c) Purchases of Equity Securities by the Issuer and Affiliated Purchasers.

We did not repurchase any of our equity securities during the quarter ended March 31, 2023.

Item 3. Defaults Upon Senior Securities.

None.

Item 4. Mine Safety Disclosures.

Not applicable.

Item 5. Other Information.

None.

Item 6. Exhibits.

(a) Exhibits

10.1+	<u>Executive Employment Agreement, dated January 1, 2023, by and between the Registrant and Ricardo C. Rodriguez (incorporated by reference to Exhibit 10.18(3) to the Company's Annual Report on Form 10-K for the year ended December 31, 2022, File No. 001-36481).</u>
10.2+	<u>Executive Employment Agreement, dated January 1, 2023, by and between the Registrant and Kelley Conte (incorporated by reference to Exhibit 10.18(4) to the Company's Annual Report on Form 10-K for the year ended December 31, 2022, File No. 001-36481).</u>
10.3+	<u>Executive Employment Agreement, dated January 1, 2023, by and between the Registrant and Gregg Landes (incorporated by reference to Exhibit 10.18(5) to the Company's Annual Report on Form 10-K for the year ended December 31, 2022, File No. 001-36481).</u>
10.4+	<u>Executive Employment Agreement, dated January 1, 2023, by and between the Registrant and Keith Schilling (incorporated by reference to Exhibit 10.18(7) to the Company's Annual Report on Form 10-K for the year ended December 31, 2022, File No. 001-36481).</u>
10.5+	<u>Executive Employment Agreement, dated January 1, 2023, by and between the Registrant and Corby Whitaker (incorporated by reference to Exhibit 10.18(8) to the Company's Annual Report on Form 10-K for the year ended December 31, 2022, File No. 001-36481).</u>
31.1	<u>Certification of principal executive officer under Section 302(a) of the Sarbanes-Oxley Act of 2002.</u>
31.2	<u>Certification of principal financial officer under Section 302(a) of the Sarbanes-Oxley Act of 2002.</u>
32	<u>Certifications of the principal executive officer and the principal financial officer under Section 906 of the Sarbanes-Oxley Act of 2002.</u>
101.INS	XBRL Instance Document - the instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document.
101.SCH	Inline XBRL Taxonomy Extension Schema Document.
101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document.
101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document.
101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document.
101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document.
104	Cover Page Interactive Data File (embedded within the Inline XBRL document).

SIGNATURES

Pursuant to the requirements of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned thereunto duly authorized.

ASPEN AEROGELS, INC.

Date: May 4, 2023

By: /s/ Donald R. Young

Donald R. Young
President and Chief Executive Officer
(principal executive officer)

Date: May 4, 2023

By: /s/ Ricardo C. Rodriguez

Ricardo C. Rodriguez
Chief Financial Officer and Treasurer
(principal financial officer and principal accounting officer)

CERTIFICATIONS UNDER SECTION 302

I, Donald R. Young, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Aspen Aerogels, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: May 4, 2023

/s/ Donald R. Young

Donald R. Young
President and Chief Executive Officer
(principal executive officer)

CERTIFICATIONS UNDER SECTION 302

I, Ricardo C. Rodriguez, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Aspen Aerogels, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: May 4, 2023

/s/ Ricardo C. Rodriguez

Ricardo C. Rodriguez

Chief Financial Officer and Treasurer (principal financial officer and principal accounting officer)

CERTIFICATIONS UNDER SECTION 906

Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002 (subsections (a) and (b) of section 1350, chapter 63 of title 18, United States Code), each of the undersigned officers of Aspen Aerogels, Inc., a Delaware corporation (the “Company”), does hereby certify, to such officer’s knowledge, that:

The Quarterly Report on Form 10-Q for the quarter ended March 31, 2023 (the “Form 10-Q”) of the Company fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934, and the information contained in the Form 10-Q fairly presents, in all material respects, the financial condition and results of operations of the Company.

Dated: May 4, 2023

/s/ Donald R. Young

Donald R. Young
President and Chief Executive Officer
(principal executive officer)

Dated: May 4, 2023

/s/ Ricardo C. Rodriguez

Ricardo C. Rodriguez
Chief Financial Officer and Treasurer
(principal financial officer and principal accounting officer)

A signed original of this written statement required by Section 906 has been provided to the Company and will be retained by the Company and furnished to the Securities and Exchange Commission or its staff upon request.
